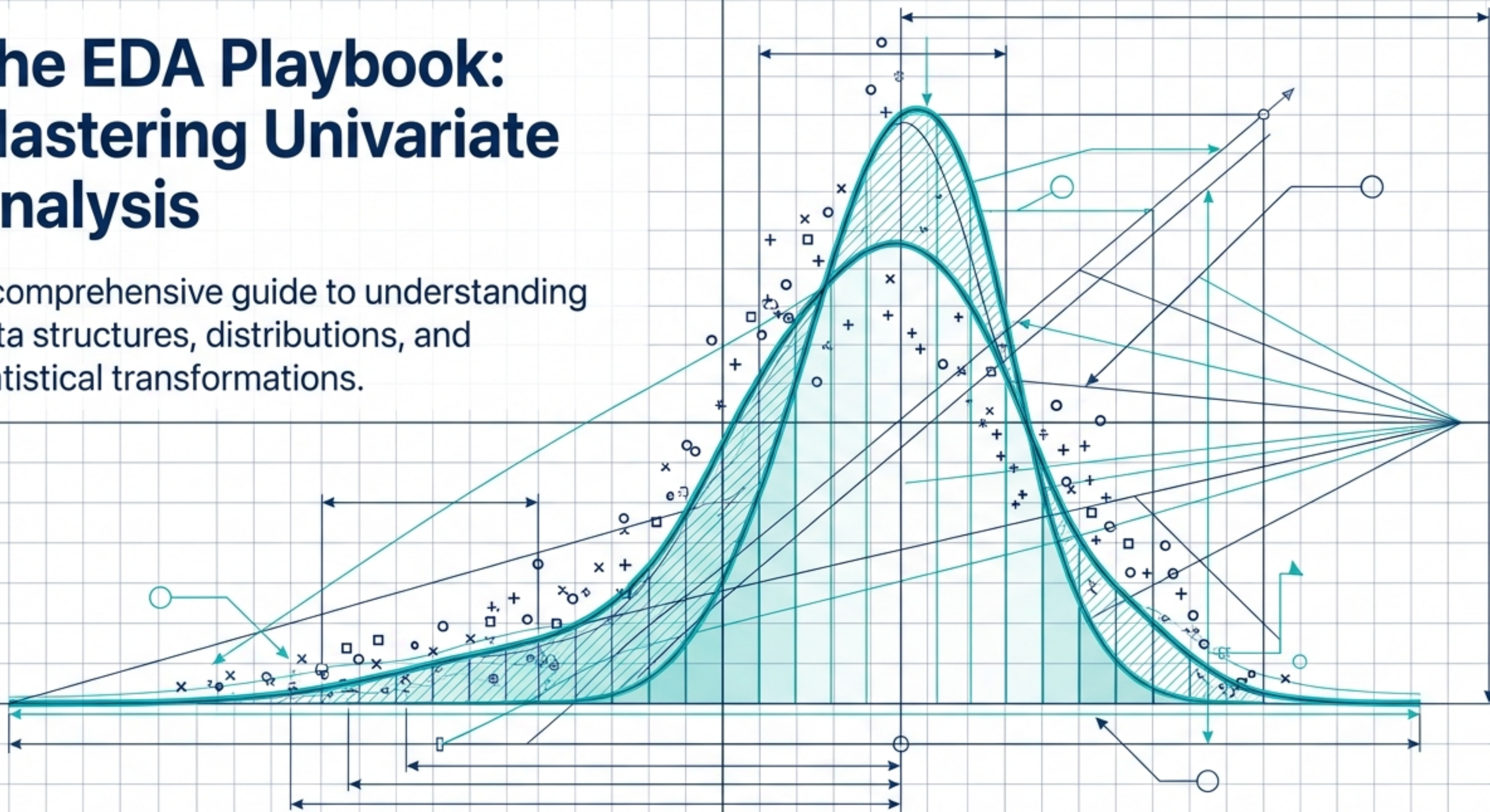
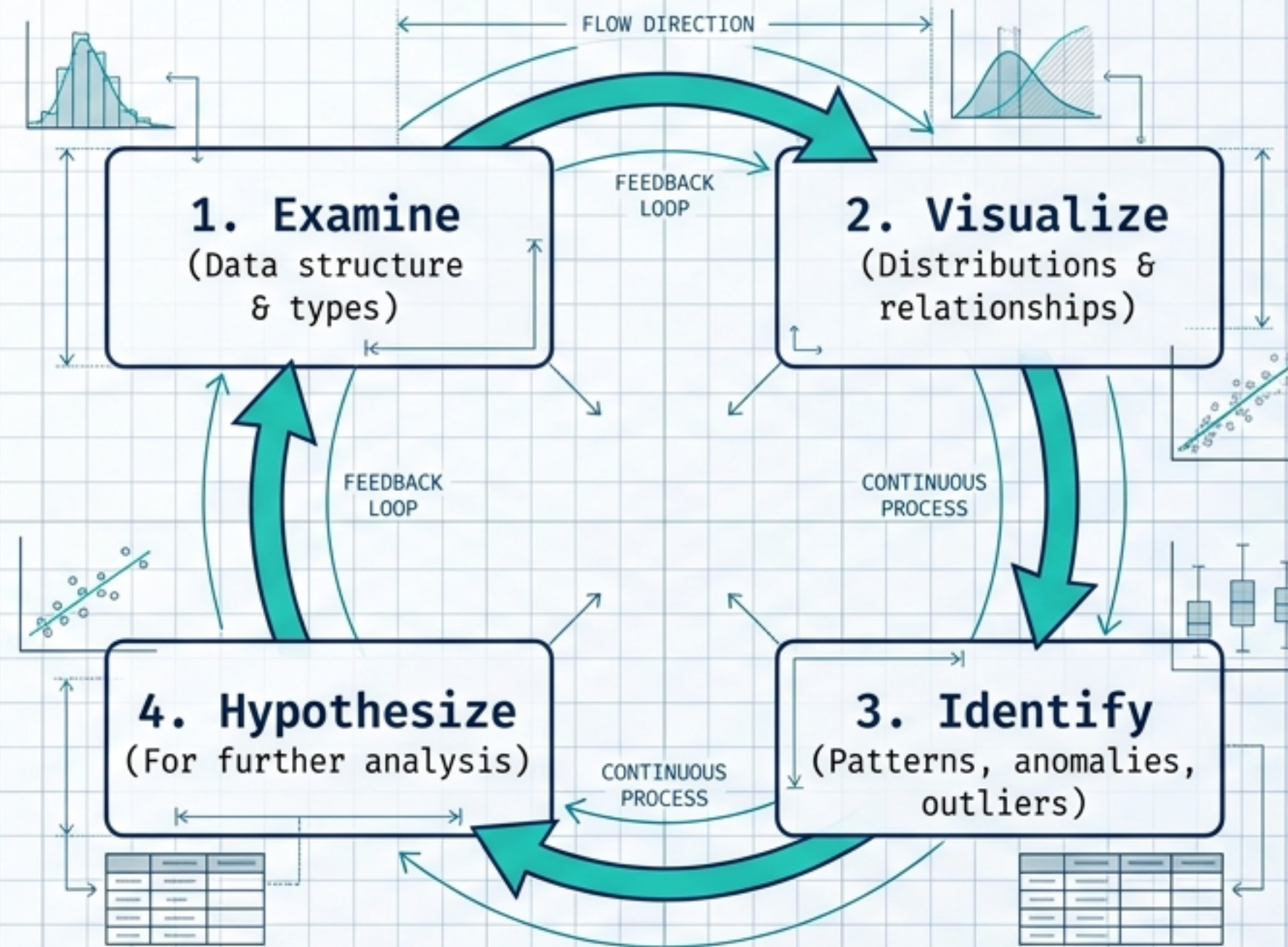


The EDA Playbook: Mastering Univariate Analysis

A comprehensive guide to understanding data structures, distributions, and statistical transformations.



The Iterative EDA Loop



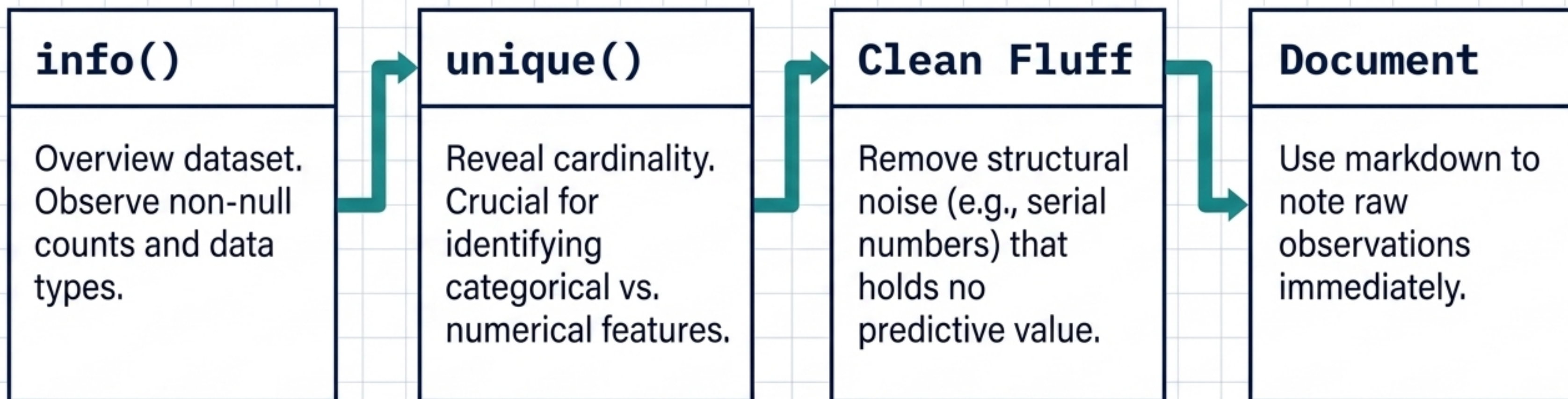
Real-World Use Case

Customer Churn Prediction: Identify which features (age, history, engagement) correlate with churn to guide model design.



THE GOLDEN RULE: Let the data speak. EDA is never about confirming pre-existing assumptions. Discover unexpected patterns that challenge initial hypotheses.

Phase 1: Initial Data Inspection



🛑 HALT BEFORE IMPUTATION. Always perform your initial analysis on the original, unmodified data distribution. Missing value imputation belongs in Phase 2.

Phase 2: Feature Engineering & Normalization Matrix

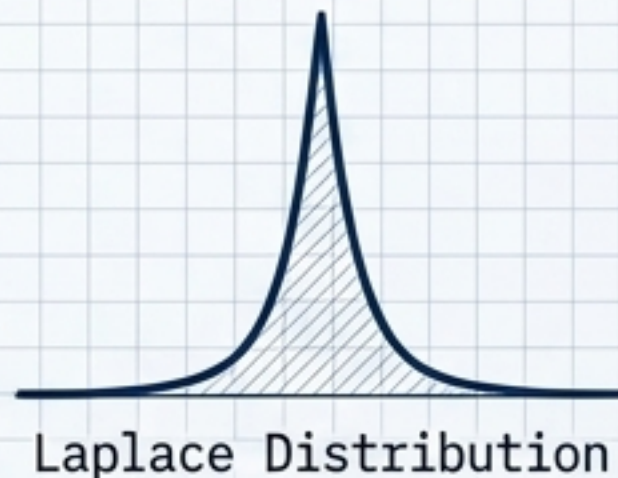
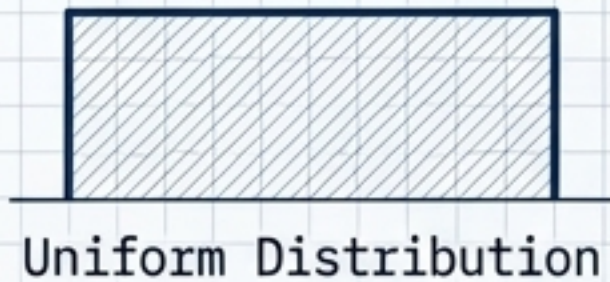
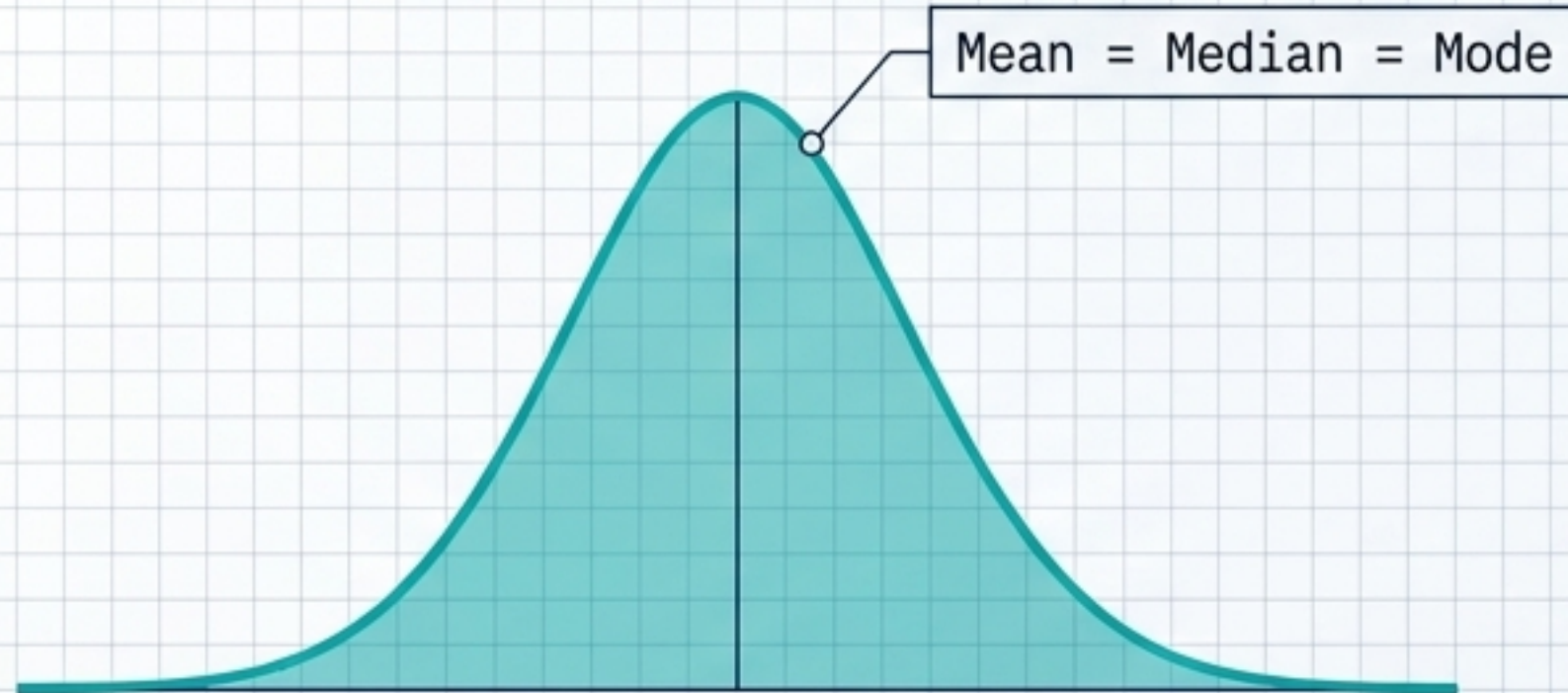
Pattern-Based	Exact Mapping	Variation Handling
Regex	Manual / Dictionary Mapping	Fuzzy Matching
Complex string cleaning.	You have a strict, known set of valid values.	Correcting typos, misspellings, and slight variations in text.



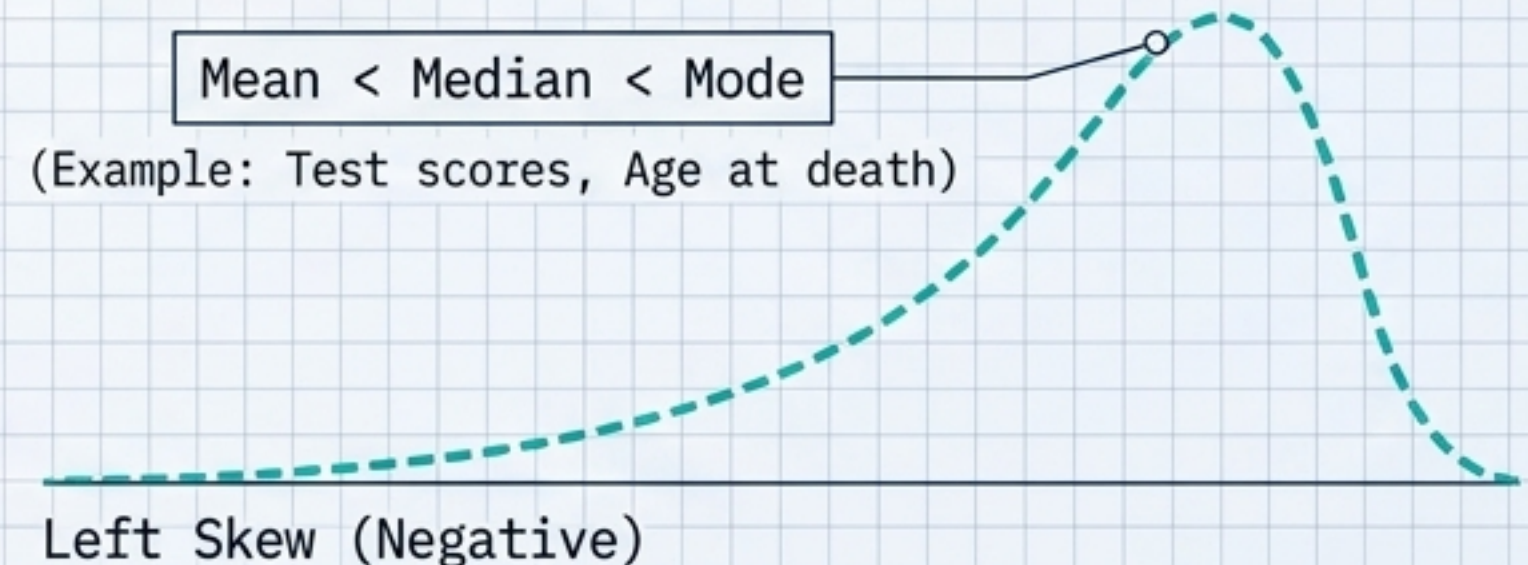
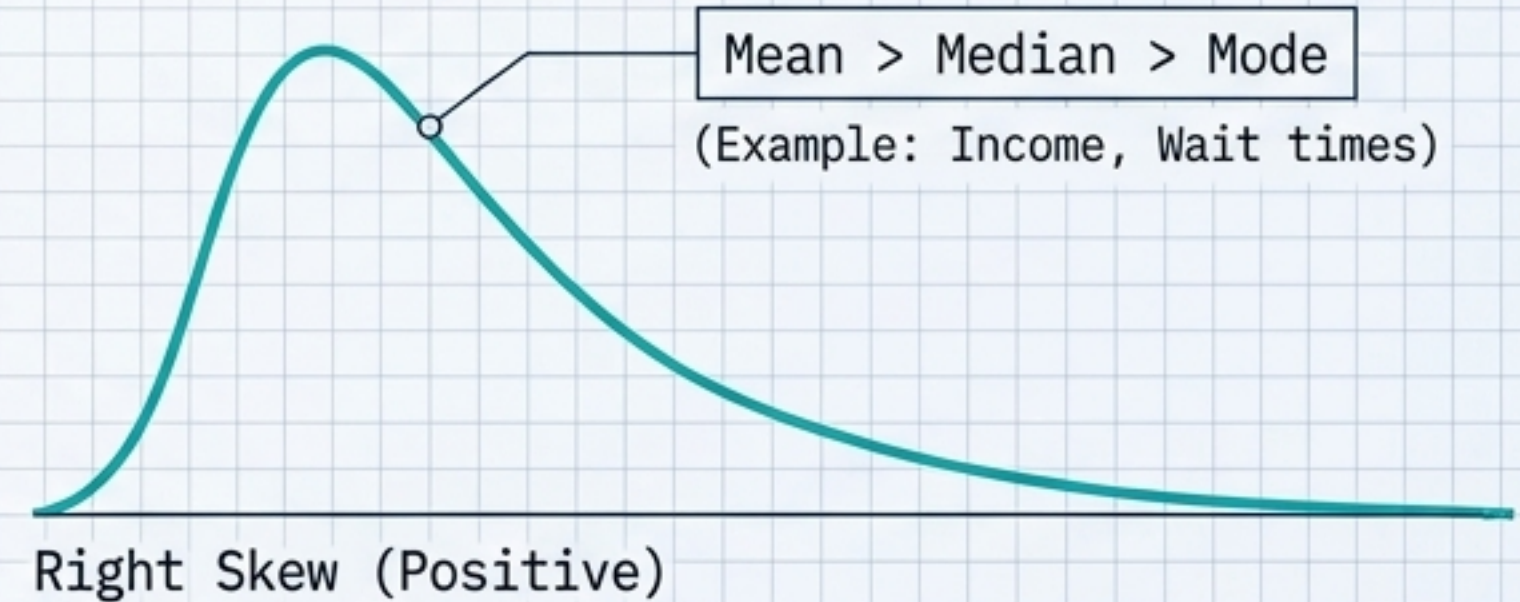
ALWAYS retain the original column when engineering features. This ensures safe comparison and provides a rollback point if the transformation fails.

The Anatomy of Distributions

Symmetric Data



Skewed Data



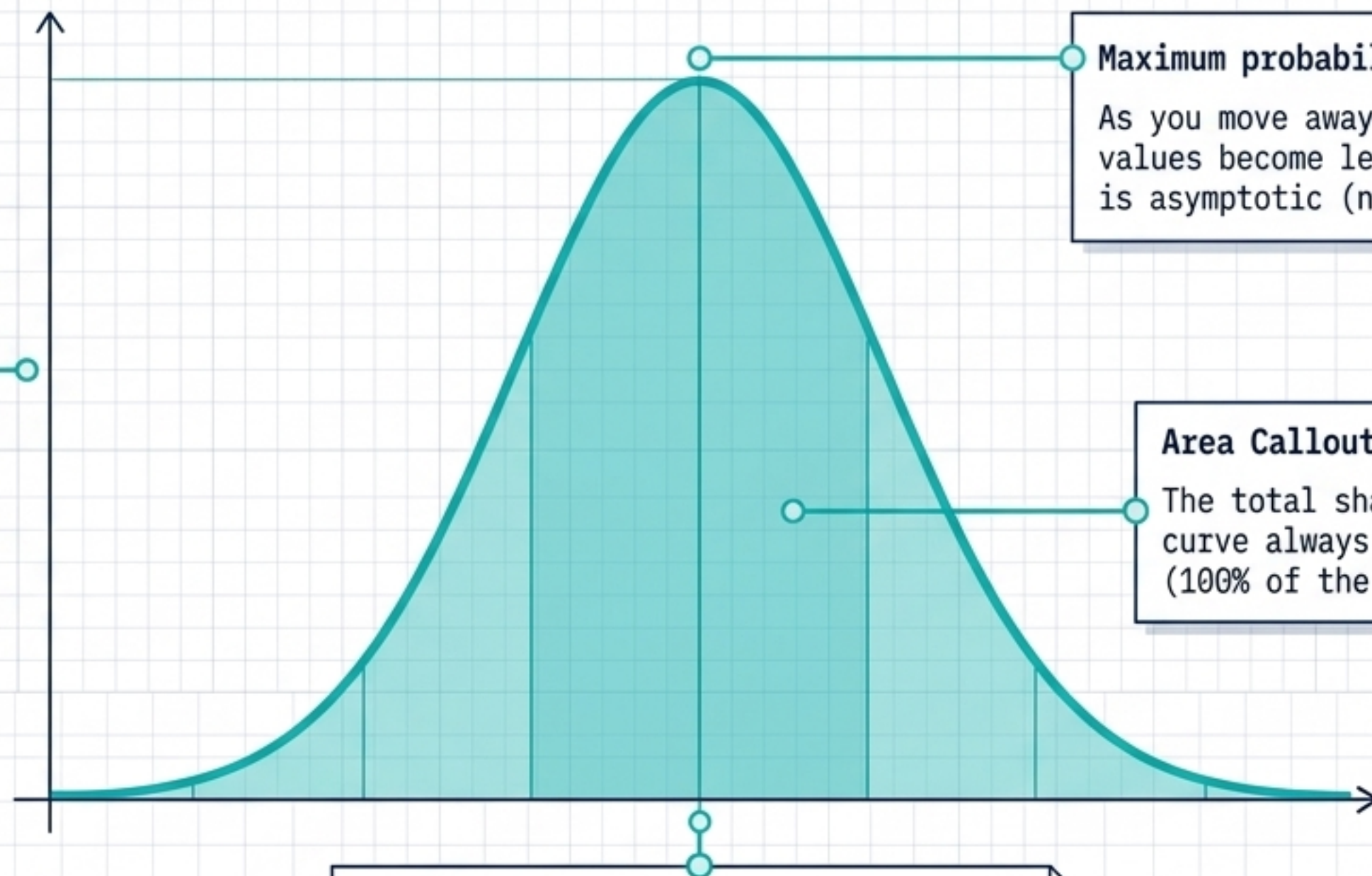
Decoding the Probability Density Curve (PDF)

Probability Density

The relative likelihood of a value occurring.

⚠ Crucial Note:

This represents relative frequency, NOT absolute count.



Maximum probability density (Mode)

As you move away from the center, values become less likely but the curve is asymptotic (never touches zero).

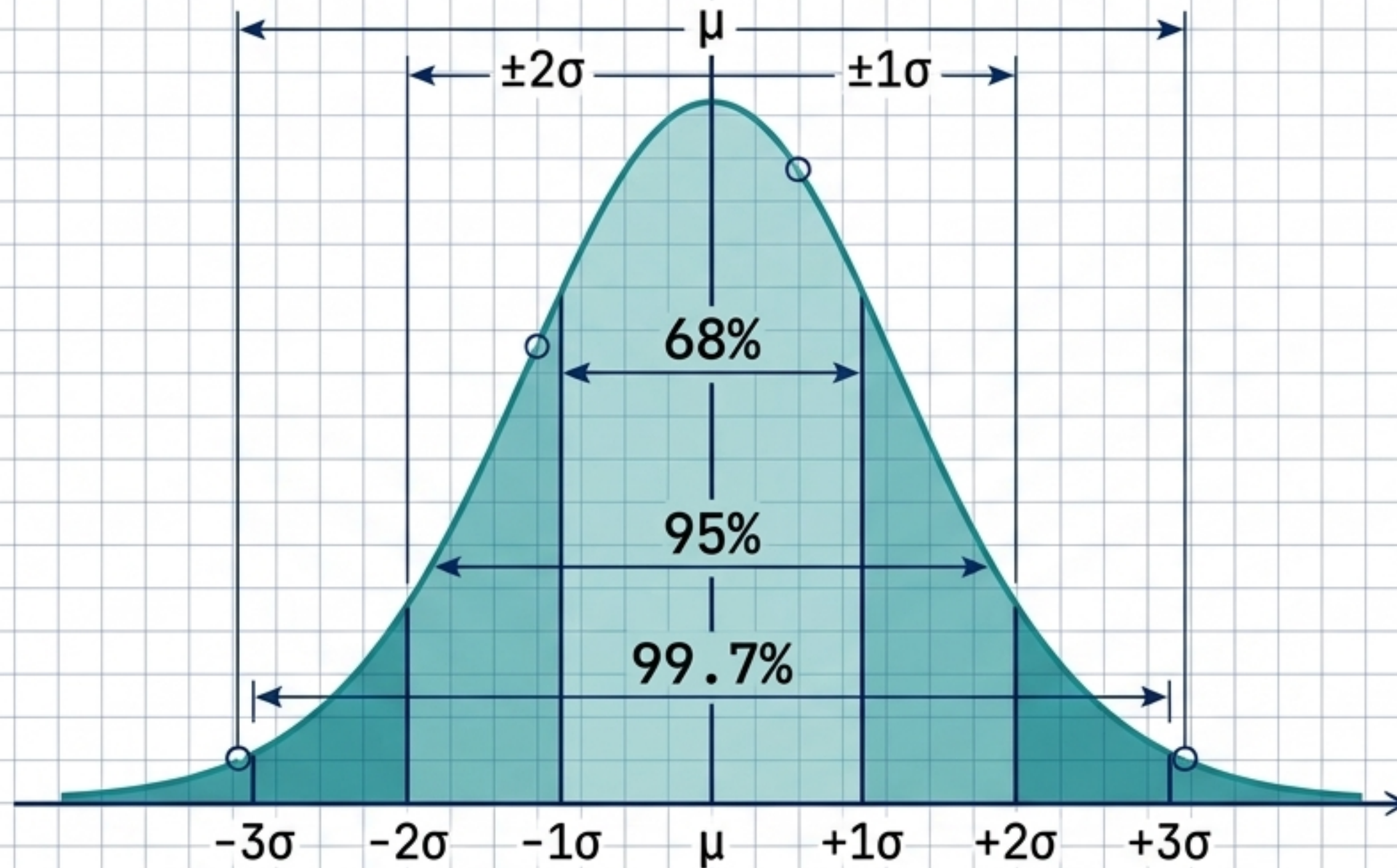
Area Callout

The total shaded area under the curve always equals exactly 1 (100% of the data).

Data Values

Represents the range of possible values (e.g., test scores, heights). The center (μ) marks the mean.

The Empirical Rule & Spread (68-95-99.7)



Mean (μ)

Represents central tendency and the "balance point".

Variance (σ^2)

$$\frac{\sum (x_i - \mu)^2}{n}$$

Squares deviations to eliminate negatives; emphasizes large deviations.

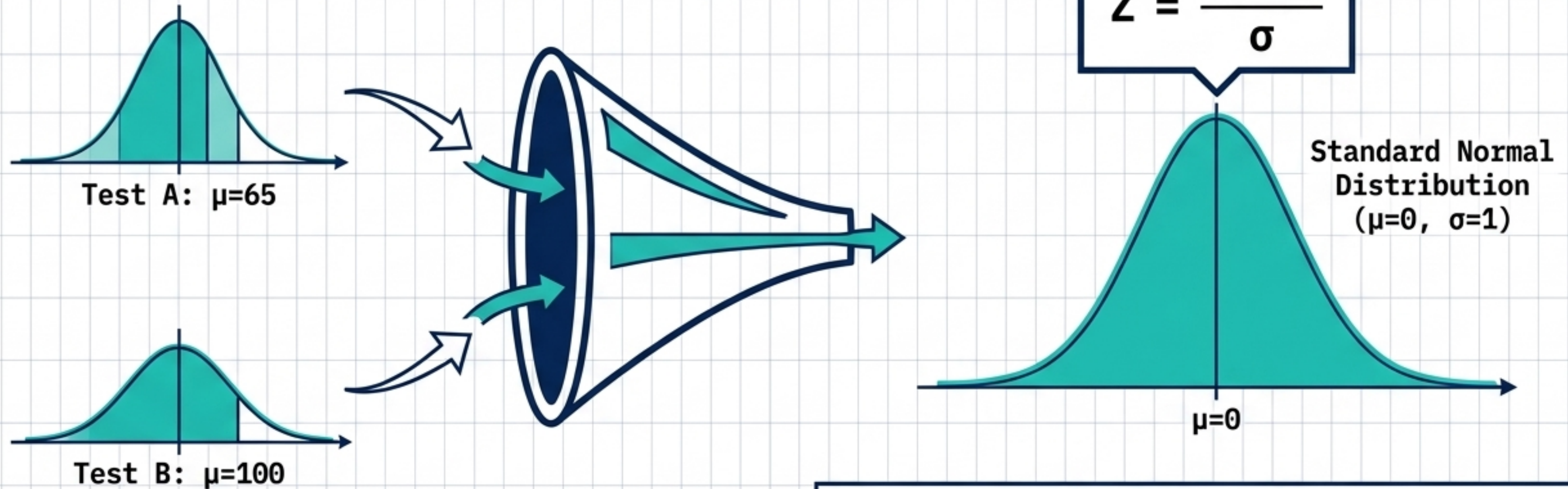
Std Dev (σ)

$$\sqrt{\text{Variance}}$$

The average distance from the mean, returned to original units.

⚠ **Population vs. Sample:** Real-world data requires sample statistics. Use Bessel's correction ($n-1$) to provide an unbiased estimate. In Python: `np.std(data, ddof=1)`.

Z-Scores: The Universal Scale



Z-Scale Map

$Z = 0$: Exactly average.

$|Z| = 1, 2, 3$: Number of standard deviations away. Beyond $\pm 3\sigma$ triggers **outlier investigation**. ⚠️

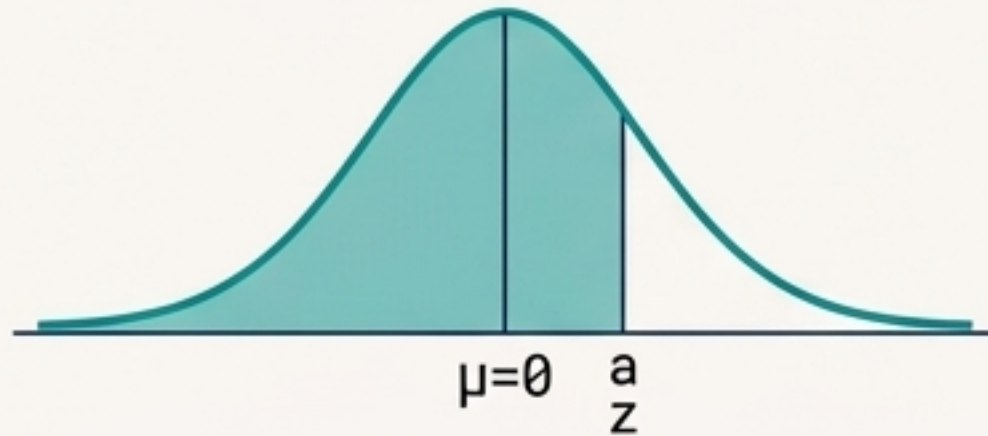
Z-Table Lookup Example & Use Cases

Z-Table Lookup Example: Finding $P(Z \leq 1.96) = 0.9750$ (97.5% of data falls below 1.96).

- **Real-World Use Cases:** Medical Testing (identifying abnormal results), Finance (comparing stock volatility).

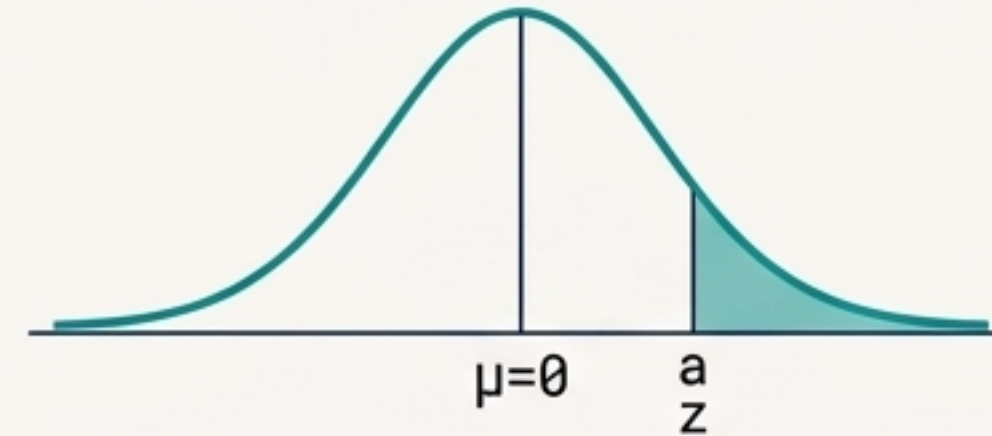
Solving Normal Distribution Questions

Less Than



$$P(X < a) = P(Z < z)$$

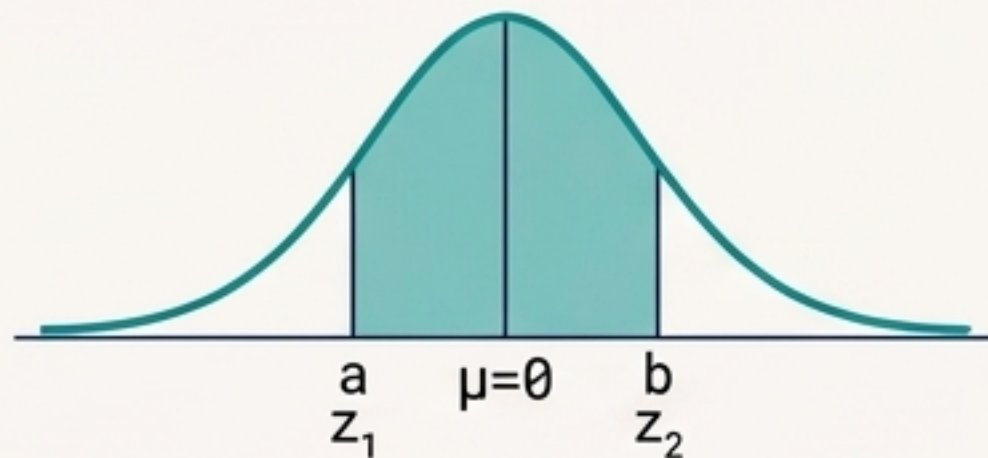
Greater Than



$$P(X > a) = 1 - P(Z < z)$$

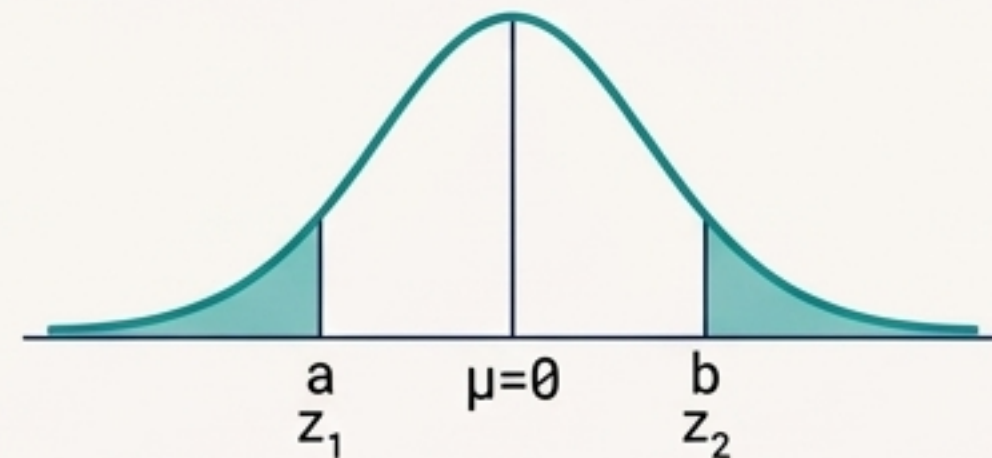
(Note: Requires complement rule!)

Between Two Values



$$P(a < X < b) = P(Z < z_2) - P(Z < z_1)$$

Outside (Tails)

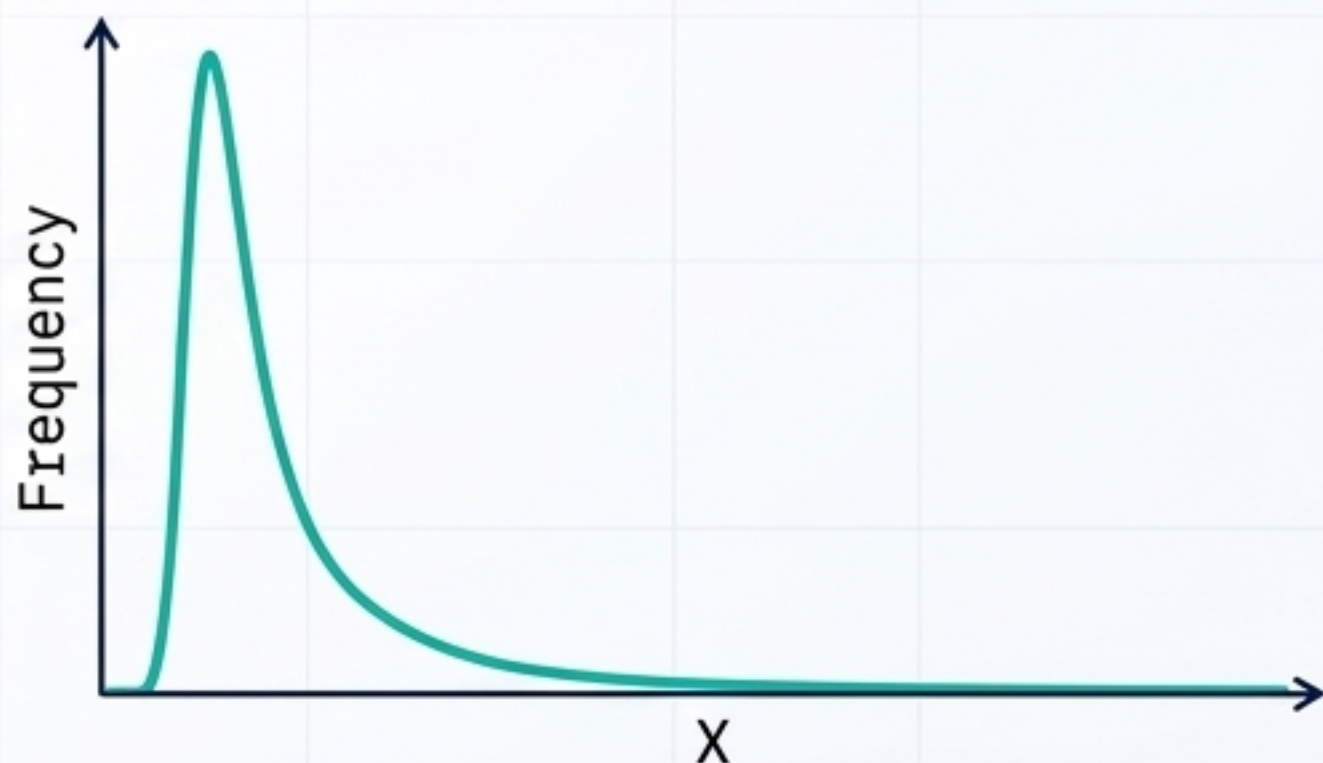


$$P(X < a) \text{ or } P(X > b) = P(Z < z_1) + (1 - P(Z < z_2))$$

The 4-Step Framework: 1. Identify Parameters → 2. Convert to Z-Score → 3. Read Z-Table → 4. Calculate Final Answer.

Diagnostic: When to Use Log-Normal Transformations

The Problem (Original Scale)

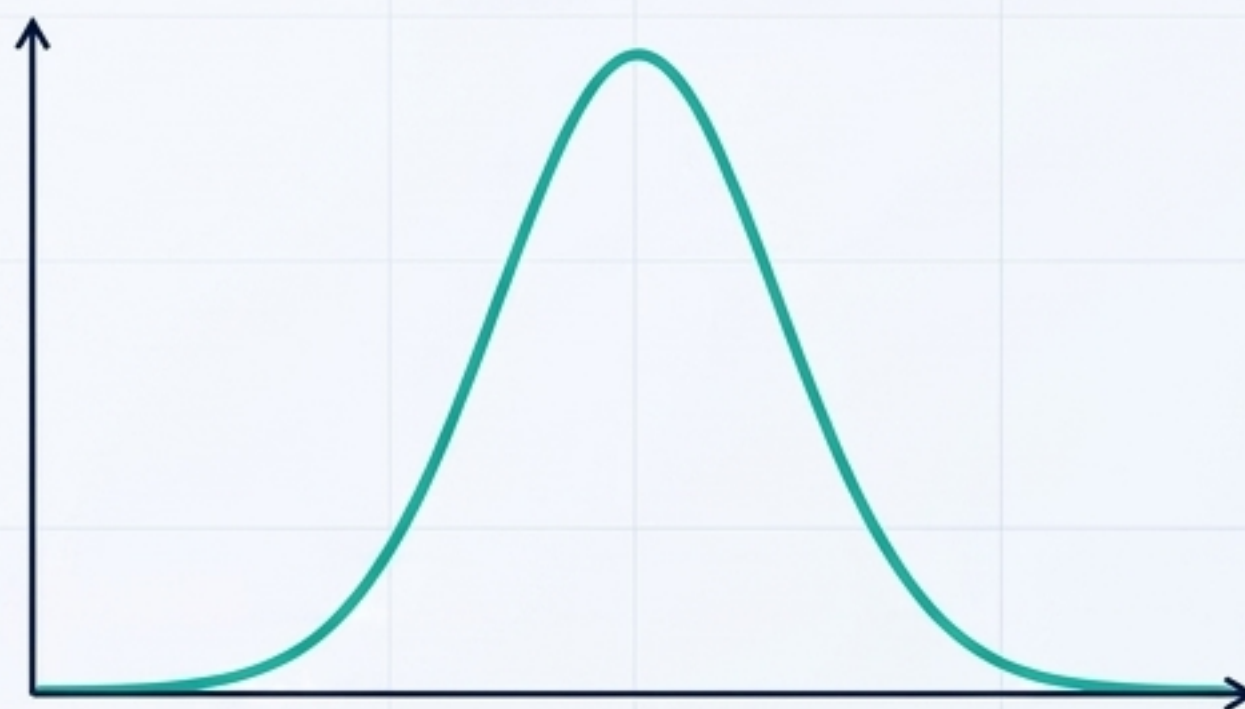


Data Traits: Only positive values, multiplicative growth. (e.g., Income, Prices, Population).

Diagnostic Issues: Outliers dominate the analysis; variance increases alongside the mean (**violating**  homoscedasticity).

Log
Transformation

The Solution (Log Scale)



Results: Compresses massive values ($1000 \rightarrow 6.9$) and expands tiny values ($10 \rightarrow 2.3$).

Stabilizes variance, reduces outlier influence, and linearizes exponential relationships.

Applying & Interpreting Log Transformations

`np.log(x)` Natural Logarithm ($\ln$)

`np.log10(x)` Log Base 10

`np.log1p(x)` CRITICAL FOR ZEROS.
Computes $\log(1+x)$ for
numerical stability.

Interpretation Box

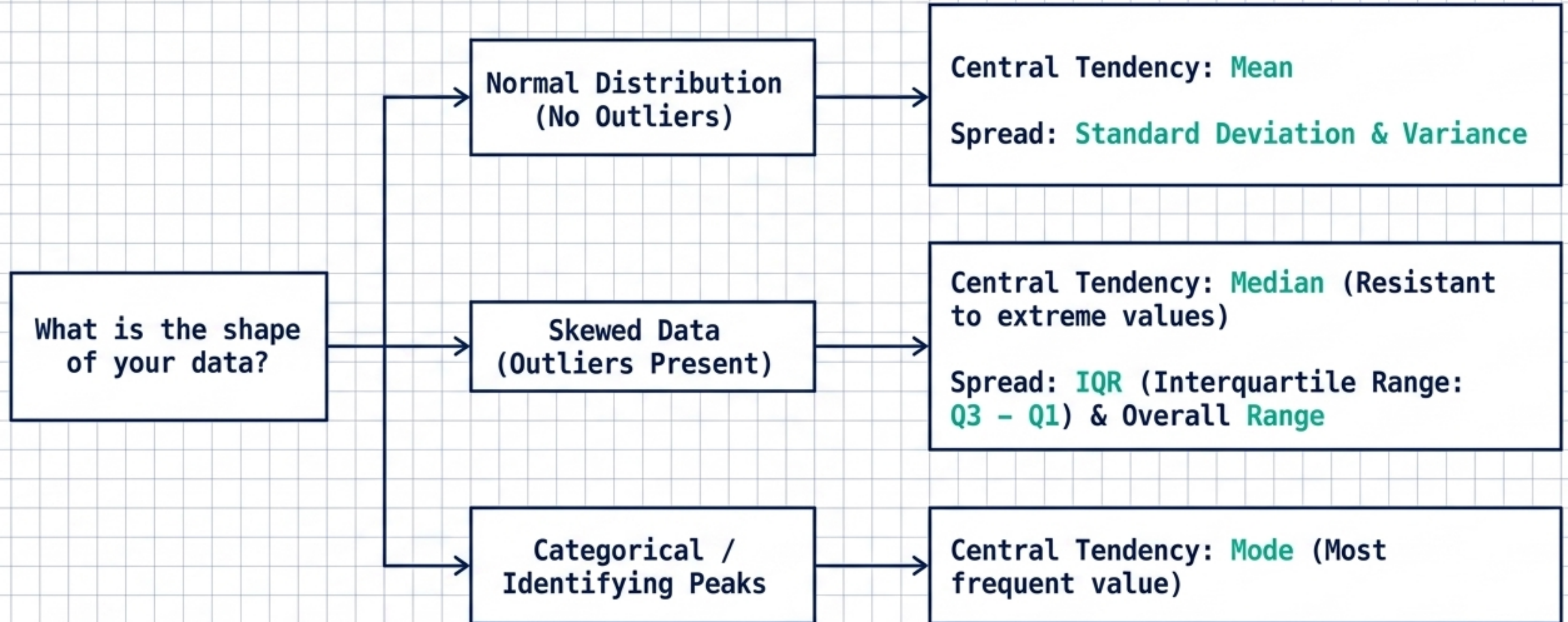
Back-Transformation: Difference in log values $\approx$ percentage change.

Example: `'log(200) - log(100) = 0.69  $\approx$  69%'` increase.

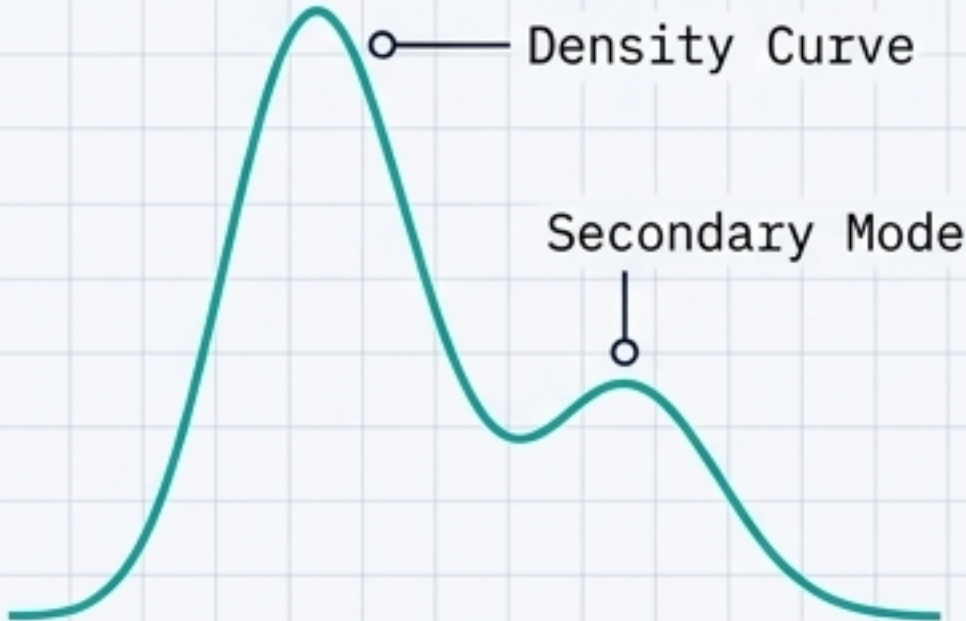
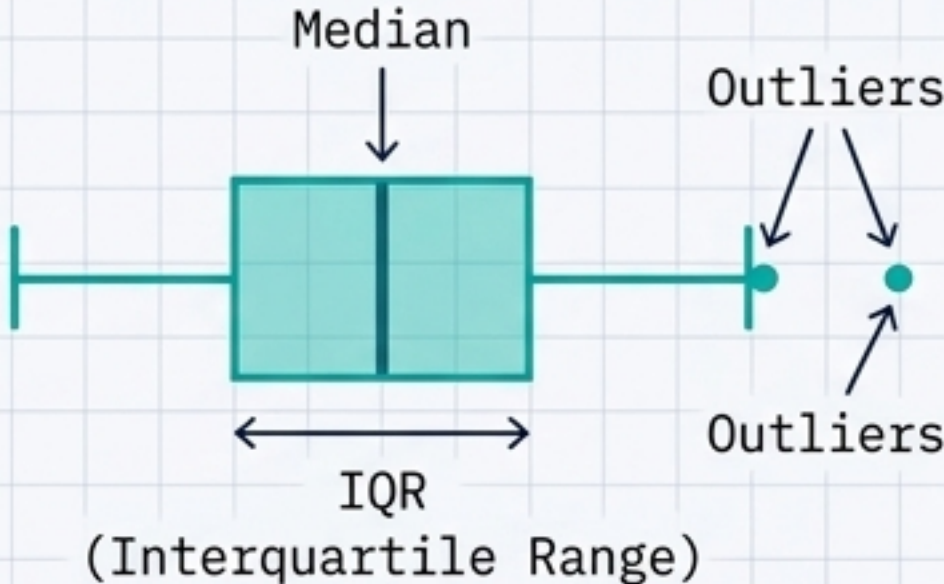
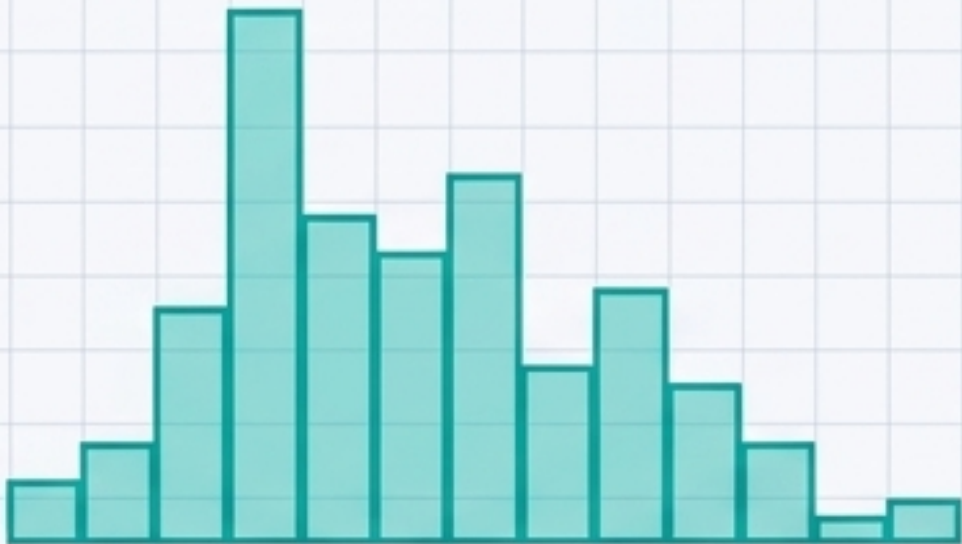
⚠ Rules of Engagement

Do **NOT** use on **negative numbers** or **left-skewed data**. Post-transformation, linear models describe **multiplicative relationships**, not additive ones.

Statistical Measures Decision Tree

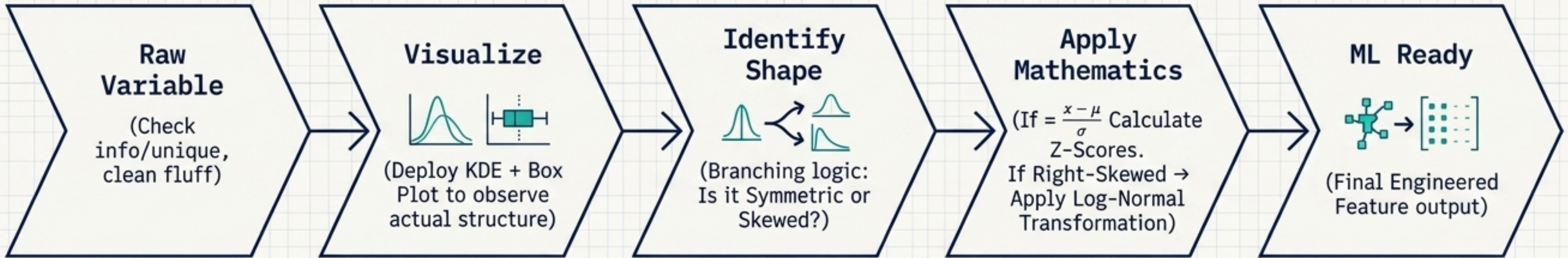


The Univariate Visualization Toolkit

KDE Plot (Kernel Density Estimation)	Box Plot	Histogram
 Best for: Understanding overall density, detecting skewness, and revealing multiple modes.	 Best for: Precise outlier detection and comparing spread/central tendency.	 Best for: Viewing actual frequency counts and discrete bins.

💡 Never rely on a single plot. Combine KDE, Box, and Histogram for a 360-degree diagnostic view.

Synthesis: The Univariate EDA Mental Model



The Data Scientist's Cheat Sheet (Summary)

Z-Scores & Spread

$$Z = \frac{X - \mu}{\sigma}$$

Empirical: 68% (1σ), 95% (2σ), 99.7% (3σ)

Key Z-Value: 1.96 = 97.5%

Distributions & Center

Symmetric: Mean = Median = Mode

Right Skew: Mean > Median > Mode

Left Skew: Mean < Median < Mode

Transformations

Right-skewed / multiplicative? Use Log.

Contains zeros? Use `np.log1p()`.

Sample vs Population

Sample Variance: $s^2 = \frac{\sum (x_i - \bar{x})^2}{(n-1)}$

Always use Bessel's correction (n-1)
for real-world sample data.